

Melvin Orichi Socana

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Infrastructure and backend engineer specializing in high-throughput LLM platforms, distributed systems, and cloud infrastructure. Production experience building scalable services and platform tooling with Python, GCP, Kubernetes, and Terraform.

EXPERIENCE

Handshake | San Francisco, CA

Jan 2025 - Present

Machine Learning & Data Platform Engineer

- Designed and productionized a self-hosted, multi-provider LLM gateway routing 30M+ tokens/min and managing \$1M+/month in inference spend, established as the required path for LLM inference across teams.
- Added autoscaling, observability, and per-client cost-attribution tooling to the gateway, enabling accurate chargeback of \$100K+/month and high-granularity visibility into LLM usage.
- Implemented FastAPI v2 real-time and asynchronous batch APIs for an internal LLM orchestration platform, unifying LLM job submission, status tracking, metadata, and BigQuery/GCS references.
- Productionized and migrated Airflow/Astronomer pipelines ingesting API data into BigQuery; adding pagination, deduplication, schema safeguards, chunked processing, and migrations to resolve failures.

Stanford NLP Research | Stanford, CA

Nov 2023 - Feb 2024

Graduate AI Research Engineer - Semantic Parsing over Wikidata

- Built a retrieval-augmented generation (RAG) parser translating questions into SPARQL, improving entity-linking performance by 20% over the prior state of the art.

Amazon | Seattle, WA

May 2023 - Aug 2023

Software Development Engineer Intern - Amazon Photos

- Developed the C#.NET backend and SQLite schema for Amazon Photos album management, including album creation, photo grouping, metadata persistence, and paginated retrieval.
- Authored a technical design documenting backend architecture, requirements, and implementation decisions for software engineering and product review.

Active Robotic Sensing Laboratory | Raleigh, NC

Aug 2021 - May 2022

Machine Learning & Robotics Research Engineer

- Developed the imaging system for BenchBot, a robotics platform for estimating plant biomass.
- Developed Python control software integrating DepthAI cameras and servos to capture machine-learning training data.

Johnson & Johnson | Santa Clara, CA

May 2021 - Aug 2021

Software/Hardware Engineer Intern

- Designed a diagnostic board for rapid validation of communication protocols, reducing test costs by 20-30%.
- Authored and executed functional test procedures for complex electronic assemblies, documenting results for engineering review.

EDUCATION

Stanford University - M.S., Electrical Engineering (Hardware & Software), 2024

North Carolina State University - B.S., Electrical Engineering & Computer Engineering, 2022

TECHNICAL SKILLS

Languages: Python, SQL, Go | Prior experience: C#, C++, TypeScript

Cloud & Infrastructure: GCP, Kubernetes, Docker, Terraform, Helm, KEDA, Datadog, Prometheus

Backend & Data: FastAPI, Airflow/Astronomer, BigQuery, Dataflow, Vertex AI, GCS, LiteLLM